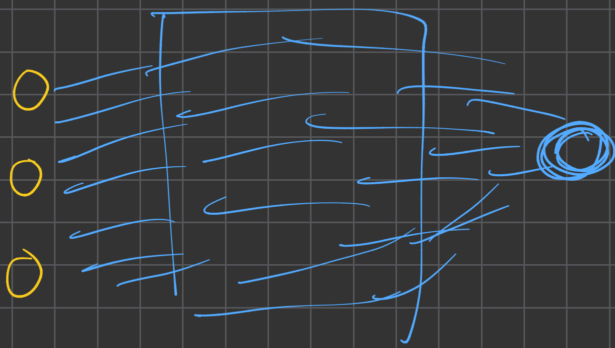
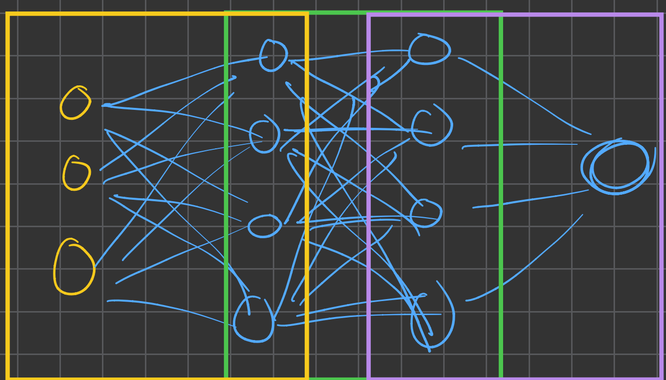
Training a Neural Network (regression)

First



My network size



(3,4) (4,4) (4,1)

(A) After forward pass, we need to calculate loss in the predictions

ys = [A, B, C, D] ypred = [a, b, c, d]

$y_2 = [A, B, C, D]$

$y\ preds = [a, b, c, d]$

$y_2 = [1, -1, -1, 1]$

$y\ preds_2 = [-0.24, -0.32, -0.92, 0.024]$

$loss = (1 - 0.024) = 0.976$] loss contr

$loss = (-1 - (-0.92)) = 0.08$] loss contr.

similarly, loss will be

$\Rightarrow |(A-a)| + |(B-b)| + |(C-c)| + |(D-d)|$

so, instead of doing modulo, we can also do square, as it also converts -ve to +ve.

$$\boxed{loss = (A-a)^2 + (B-b)^2 + (C-c)^2 + (D-d)^2}$$

this is called MSE loss

so, $\boxed{loss > 0}$ ✗

Made with Goodnotes

General formula (MSE)

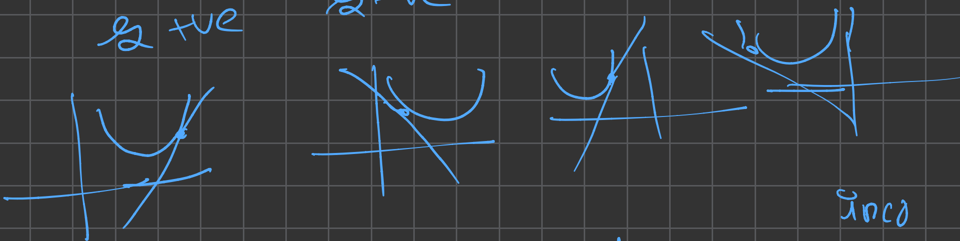
$$MSE = \sum_{i=0}^{n} \left((y(i) - ypred(i)) ** 2 \right)$$

Convention L = +ve

w +ve

w -ve

s +ve   s -ve   s +ve   s -ve



decr (w)   incr. (w)   decr (w)   incr (w)

As per obs., opp. of dir^n of slope, adjust your parameters

w.data += $(-\alpha) *$ slope (→ w.grad)